

Introduction to Statistical Methods

(S1-25 AIMLCZC418) – Assignment 1

AIML Section- 1

Each question carries 2.5 Marks (2.5 x 4 = 10 Marks)

Duration: 20th Nov, 2025 – 2nd December 2025

1) Submissions are individual

2) Solve these on paper, scan, and upload

3) Plagiarism results in zero marks

4) Write your name, BITS ID and Section on each page

5) Only handwritten solutions with formula, full steps with proper justification are required.

Q1. A company's monthly maintenance costs (in thousands) for the last year are:

18,20,19,21,23,20,22,19,18,20,21,120

- Find the mean, median, and mode. Identify whether the data is left/right skewed.
- Give the five-point summary and IQR
- Suggest which measure (mean/median) should be used for decision-making and justify.

Q2. A company's cybersecurity system tracks two potential events on its network: a direct intrusion attempt and the presence of malware (detected by a separate scan). Based on the system's reports over a specific period, the probability of an intrusion attempt occurring is 0.10. The probability of malware being detected on the network is 0.08. Furthermore, the probability that the system experiences both an intrusion attempt and a malware detection simultaneously is 0.03. Find

- Probability of intrusion or malware
- Probability of intrusion but no malware
- Probability of neither intrusion nor malware

Q3. A company has three plants to manufacture Electrical cars. Plant A manufactures 30%, Plant B manufactures 45% and Plant C manufactures 25%. At Plant A 98% are rated standard quality, at Plant B, 97% are rated standard quality and at Plant C, 98% are rated standard quality.

- What is the probability that electrical cars selected at random and came from Plant A, if it is known that the car is of standard quality?
- What is the probability that electrical cars selected at random and came from Plant C, if it is known that the car is of standard quality?
- What is the probability that the car selected at random is of not a standard quality?

Q4. A man has a biased coin where the probability of getting a Head (H) is $\frac{2}{3}$. He also has a standard deck of 52 playing cards. He flips the coin.

- If the result is Heads (H), he draws a single card from the deck and reports the suit correctly (i.e., truthfully) with a probability of $\frac{4}{5}$.
- If the result is Tails (T), he draws a single card from the deck and lies about the suit with a probability of $\frac{4}{5}$. The lie is chosen equally from the other three suits.

The man performs the experiment and reports that the card drawn was a Diamond. Find the probability that the coin flip result was Heads (H).

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AIML Section- 2

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Q1. A manufacturing unit recorded the number of defective items produced per month over a year as follows: 12,15,14,13,16,18,14,15,13,17,14,56

- Compute the mean, median, and mode of the defects. Comment on whether the distribution is symmetrical or skewed.
- Give the five-point summary and find the outliers if any using IQR method.
- The company wants to report monthly performance to clients. Should they report mean or median? Justify mathematically.

Q2. In a class, 70% of students attended the morning session, 55% attended the afternoon session, 40% attended both. Draw a Venn diagram for clarity and find the probability that a randomly chosen student:

- Attended at least one session
- Attended exactly one session
- Attended neither session

Q3. A company wants to predict whether a website visitor will Subscribe to their newsletter based on one feature: Device Type. You are given the following training data:

Visitor	Device	Subscribe
1	Mobile	No
2	Mobile	Yes
3	Desktop	Yes
4	Desktop	No
5	Tablet	Yes

- Compute the prior probabilities: $P(\text{Subscribe} = \text{Yes})$, $P(\text{Subscribe} = \text{No})$
- A new visitor is using a Mobile device. Using the Naive Bayes classifier, predict whether they will Subscribe.

Q4. A binary classifier is trained to classify emails as spam and ham (not spam). Suppose the classifier is 89% accurate i.e., if an email is spam the classifier will mark it as spam with probability 0.89 and if it is not spam(ham) it will mark it as not spam with probability 0.89. A person is chosen to test this classifier: 24% of the emails received by this person are spam.

- What is the probability of classifying an email as not spam?
- A mail is randomly selected from the inbox and fed to the classifier. What is the probability that an email that is classified as spam is actually spam?

----ALL THE BEST----

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(S1-25 AIMLCZC418) – Assignment 1

AIML Section- 3

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Q1. A server recorded latency (in milliseconds) of a distributed system over 15 trials:

32,30,29,31,600,33,30,29,28,31,30,29,32,31,30

- a) Calculate mean, median, and mode and determine the type of skewness using these values.
- b) Give the five-point summary and IQR
- c) For evaluating server performance stability, which measure of central tendency is most appropriate, and why?

Q2. A data analytics company uses two different machine-learning models to make predictions.

- **Model A** is used 55% of the time. When it is used, it makes a correct prediction 80% of the time.
- **Model B** is used 45% of the time. When it is used, it makes a correct prediction 65% of the time.

If a prediction is selected at random, what is the probability that the prediction is correct?

Q3. A company wants to predict whether a customer will Buy a product based on a single feature: Age Group. You are given the following training dataset:

Customer	Age Group	Buy
1	Young	No
2	Young	Yes
3	Middle	Yes
4	Senior	Yes
5	Senior	No

A new customer belongs to the Senior age group. Using Naive Bayes Classifier, predict whether they will Buy.

Q4. A multiple-choice exam has 4 choices for each question. A student has studied enough so that the probability they will know the answer to a question is 0.5, the probability that they will be able to eliminate one choice is 0.25, otherwise all 4 choices seem equally plausible. If they know the answer they will get the question right. If not, they must guess from the 3 or 4 choices. As a teacher you want the test to measure what the student knows. If the student answers a question correctly what is the probability, they knew the answer?

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(S1-25 AIMLCZC418) – Assignment 1

AIML Section- 4

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Q1. A librarian records the number of pages read by 12 students during a 20-minute silent reading session.

The data (in number of pages) are: 16, 20, 12, 25, 18, 22, 19, 15, 13, 21, 17, 28.

a) Compute the mean, median, sample variance, range, and Q1, Q3 for this group.

b) Use the IQR method to identify the outliers in the data set.

Q2. A student takes his examination in four subjects A, B, C, and D. He estimates his chance of passing in A as $\frac{4}{5}$, in B as $\frac{3}{4}$, in C as $\frac{5}{6}$ and in D as $\frac{2}{3}$. To qualify, he must pass in A and at least two other subjects. What is the probability that he qualifies?

Q3. You are given the following training dataset:

Email	Sender	Contains Link	Spam
1	Friend	No	No
2	Unknown	Yes	Yes
3	Friend	Yes	No
4	Unknown	No	No
5	Unknown	Yes	Yes

Using Naive Bayes with Laplace smoothing, predict whether a new email is Spam if it has:

Sender = 'Unknown' and Contains Link = 'No'

Q4. Given the table below, consider the following random experiment:

Capacitance (μF)	Box #			Total
	1	2	3	
0.1	35	25	40	100
0.5	75	95	70	240
1.0	60	10	65	135
Total	170	130	175	475

Assume that the box selection and the capacitor selection are both with equal probability.

(i) If a capacitor of $0.1 \mu F$ is selected, what is the probability that it came from box 3?

(ii) If box 2 is selected, what is the probability that it has a capacitor of $0.5 \mu F$?

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AIML Section- 5

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Q1. The daily step count (in thousands) for 14 employees over a week are:

8.2, 10.0, 9.5, 12.1, 11.3, 8.8, 9.0, 7.7, 12.8, 10.6, 9.4, 8.1, 13.5, 11.8.

- (i) Calculate the mean, median, sample variance, SD, range, and interquartile range.
- (ii) Identify whether the data is left/right skewed.
- (iii) List any data points considered as outliers.

Q2. An engineering college has 120 students: 45 in Computer Science (20 scholarship holders), 35 in Electrical (15 scholarship holders), and 40 in Mechanical (10 scholarship holders). Five students hold double major, with 2 receiving scholarships, counted in both departments. If a student is selected at random, find the probability that the student is either from Electrical or is a scholarship holder.

Q3. You have the following dataset:

Patient	Fatigue	Loss of Smell	Sore Eyes	Has Flu
P1	Yes	Yes	No	Yes
P2	No	No	Yes	No
P3	Yes	No	No	Yes
P4	No	Yes	Yes	No

A new patient exhibits: Fatigue = No, Loss of Smell = Yes, Sore Eyes = No

Apply Naive Bayes Classifier to predict if the patient has flu.

Q4. An investor made 40% of investment in stocks, 20% in mutual funds and the rest in personal business. There are 20%, 10% and 15% of chances of obtaining profit in stocks, mutual funds and personal business respectively.

- a. What is the probability that the investor gets profit?
- b. Given that the investor got some profit, what are the probabilities that he got from stocks, mutual funds and personal business?

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(S1-25 AIMLCZC418) – Assignment 1

AIML Section- 6

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Q1. Temperatures (in °C) measured at noon for 11 consecutive days are:

31, 29, 27, 34, 32, 28, 33, 29, 35, 26, 30.

(i) Compute the mean, median, variance, range, and range, and Q1, Q3 for this group.

(ii) Identify whether the data is left/right skewed.

(iii) Use the IQR method to identify the outliers in the data set.

Q2. In a class of 40 students, there are 18 boys and 22 girls. Out of the boys, 7 participate in sports and 6 scored an 'A' grade, with 3 boys involved in both. Among the girls, 9 participate in sports, 8 scored an 'A' grade, and 4 girls did both. If a student is picked at random, what is the probability the student is either involved in sports or scored an 'A' grade?

Q3. You have the following dataset:

Person	Trouble Sleeping	Low Energy	Anxiety	Has Depression
A1	Yes	Yes	Yes	Yes
A2	No	Yes	No	No
A3	Yes	No	Yes	Yes
A4	No	No	No	No

A new person arrives with: Trouble Sleeping = Yes, Low Energy = No, Anxiety = Yes.

Predict whether the person has depression using Naive Bayes Classifier.

Q4. In a university, 5% of engineering students and 2% of arts students score above 95% in the final exam. Additionally, 70% of the students are enrolled in arts. If a randomly selected student score above 95%, what is the probability that the student is an arts student?

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AIML Section- 7

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Q1. A group of 12 students obtained the following scores in a mathematics quiz:

28, 35, 40, 42, 45, 50, 52, 55, 60, 65, 70, 75.

(i) Calculate the mean, median, variance, SD, range, and interquartile range.

(ii) Identify whether the data is left/right skewed.

(iii) Using the IQR method, identify whether any outliers exist in the dataset, with justification.

Q2. A patient undergoes two independent diagnostic tests for the same disease. Each test has a 90% sensitivity and 95% specificity.

a) What is the probability that both tests return positive for a person who has the disease?

b) What is the probability that both tests return positive for a person who does not have the disease?

Q3. You have the following dataset:

ID	Plan	Support Calls	Churn
1	Basic	Low	No
2	Premium	High	Yes
3	Basic	High	Yes
4	Premium	Low	No
5	Basic	Low	No

Using Naive Bayes, predict Churn for a new customer with: Plan = Premium, and Support Calls = High.

Q4. An e-commerce company uses an AI model to automatically classify newly uploaded products into categories. About 20% of all products on the platform are truly Electronics items. The AI model is not perfect:

- If a product is actually an Electronics item, the model correctly labels it as Electronics 93% of the time.
- If a product is *not* Electronics, the model still mistakenly labels it as Electronics 4% of the time.

A new product has just been uploaded, and the model has labelled it as Electronics.

What is the probability that the product is an Electronics item?

----ALL THE BEST----

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AIML Section- 8

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Q1. The completion times (in days) for 12 recent software modules are:

14, 16, 15, 18, 17, 16, 15, 14, 17, 16, 15, 72

- Compute the mean, median, sample variance, range, and Q1, Q3 for this group.
- Determine the skewness (left/right) based on central tendencies.
- Recommend which measure the project manager should report to clients and justify why.

Q2. Among 48 research scholars, 20 collaborated with other institutes, 18 participated in international conferences, and 8 did both. If a scholar is picked at random, what is the probability they collaborated with other institutes or participated in a conference?

Q3. You have the following dataset:

ID	Sender	Has Attachment	Spam
1	Friend	No	No
2	Unknown	Yes	Yes
3	Friend	Yes	No
4	Unknown	No	No
5	Unknown	Yes	Yes

Apply Naive Bayes to predict Spam for an email with: Sender = Unknown, Has Attachment = No.

Q4. A factory uses an IoT monitoring system to detect abnormal sensor readings in its machines.

Actual anomalies are rare — only 1% of all readings come from truly abnormal machine behaviour.

The anomaly-detection system works as follows:

- When a real anomaly occurs, the system correctly flags it 88% of the time.
- When everything is normal, the system still raises a false alarm 6% of the time.

One of the latest sensor readings has been flagged as anomalous by the system.

What is the probability that the reading is actually from a real anomaly?

----ALL THE BEST----